## Trees and climate change

Forests have been especially hard-hit, mainly due to the dual demands of lumber required for construction and fuel, and land cleared for agriculture and development. An estimated 54,000 sq miles (140,000 sq km) of tropical rain forest—which contains the greatest biodiversity—is cleared each year. Scientists will never know how many forest-dwelling species died out before they were “discovered.”

Deforestation also contributes to global climate change. As trees photosynthesize, they absorb carbon dioxide and release oxygen. However, less forest means that more $CO_2$ stays in the atmosphere, fueling the greenhouse effect and global warming.

Carbon and other greenhouse gases are emitted from cars and factories burning fossil fuels. Since 1958, American scientist Charles Keeling’s measurements of atmospheric $CO_2$ have shown that $CO_2$ emissions are increasing at an ever-faster rate. While a minority of scientists maintain that human activity is not responsible, climate change has warmed the continents. The consequences, including trees coming into leaf and flowers blooming earlier in spring, may benefit some organisms but could prove disastrous for others.

## Toxic controls

The introduction of pesticides, such as DDT, to increase crop harvests proved to be an environmental disaster: they eradicated useful invertebrates as well as harmful ones; they caused cancers in humans; and rendered birds of prey infertile. Rachel Carson’s 1962 book *Silent Spring* highlighted many of these issues, and caused a partial rethink of pesticide use. The work of several other ecologists has resulted in legislative controls to mitigate the environmental impact.